

Analysis and Design of Bang-Bang PD-Based Phase Noise Filter

Tongning Hu, *Student Member, IEEE*, Shilei Hao, *Student Member, IEEE*,
and Qun Jane Gu, *Senior Member, IEEE*

Abstract—In this paper, we present a bang-bang phase detector (BBPD) and a delay-line frequency discriminator-based phase noise filter (PNF). With a larger phase detection gain, the BBPD-based PNF enhances the sensitivity by suppressing the charge pump noise. A time-amplifier and a five times voting machine are introduced together with the sense-amplifier-flip-flop to minimize the BBPD noise to make its effects negligible in the PNF sensitivity. At 1-MHz offset, the PNF demonstrates 15-dB phase noise suppression with -120.2-dBc/Hz sensitivity. Its phase noise suppression offset frequency is from 100 kHz to 8 MHz with 10–10.1-GHz input frequency range. The circuit is fabricated in a 65-nm CMOS technology, and dissipates 98-mW power.

Index Terms—Bang-bang phase detector (BBPD), delay line, frequency discriminator (FD), phase noise filter (PNF), phase noise suppression, phase noise cancellation, time-amplifier (TA), voting machine.

I. INTRODUCTION

LOCAL oscillators (LOs) play a key role in a variety of high performance systems, such as wideband communications, high resolution imaging and radar. Generally, phase noise is a limiting factor for the LOs. In essence, phase noise improvement techniques can be categorized in two different classes: Direct phase noise improvement in frequency synthesizers or phase-locked loops (PLLs) [1]–[4], and phase noise filter/cancellation for the input LO signals [5], [6]. In the former case, all the approaches rely on a clean reference clock. To the contrary, in the latter case, a frequency discriminator (FD) is employed to extract the phase noise information without the need of a reference clock. Furthermore, the phase noise filter/cancellation structure is independent of the LO generation. A variety of studies have investigated in the latter technique. In [5], a FD is performed to cancel the out-of-band phase noise in a ring-oscillator based PLL with a feed-forward approach, which features wide suppression bandwidth. Reference [6] adopts a FBAR in the phase demodulator to suppress the phase noise of the voltage-controlled-oscillator (VCO). Its input frequency range is limited by the high Q FBAR, which is

below 5 MHz. As indicated in [5] and [6], the mixer based FD needs quadrature-phase inputs for phase detection, thus requiring complicated phase shifting circuits. Moreover, the mixer based FD is susceptible to the amplitude noise because of its voltage multiplication nature. Recently, a PFD/CP based FD is proposed in [7] and [8]. This approach enables in-phase inputs and is insensitive to the amplitude noise. However, the small PFD gain ($1/2\pi$) would amplify charge pump (CP) noise, which is the limiting factor of the system sensitivity. In addition, a subsequent baseband amplifier with a large gain is required to boost the loop gain, which introduces extra noise and consumes more power.

In this paper, we propose a BBPD and delay line FD based phase noise filter (PNF) to mitigate the above issues. Compared to the other FDs, this design replaces the mixer and linear PFD with a BBPD. This choice leads to in-phase input requirement and a better PNF noise floor, since the BBPD with extremely high gain suppresses the CP noise. This PNF acts as a general function unit to improve the clock phase noise without the need of a superior reference. Moreover, the PNF can be employed into an array system to further improve the phase noise due to the uncorrelated noise feature among each PNF [9]. As analyzed in [9], the phase noise sensitivity improves by $10\log(N)$ in the N-element array system.

This paper is an expanded version of [11] by presenting the details of the theoretical analysis, design optimization, circuit implementation and measurement results. Besides the quantization noise, BBPD itself also introduces random noise to the system. Therefore, a time-amplifier (TA) and a voting machine are added in this expanded version to address the BBPD noise issue in [11]. It is organized as follows. Section II compares the current phase noise measurement techniques and introduces the proposed BBPD based PNF. Section III analyzes the limit cycle regime and random noise regime for the PNF. Section IV discusses the implementation of the PNF. The measurement results are shown in Section V and the conclusion is given in Section VI.

II. PHASE NOISE MEASUREMENT TECHNIQUES AND PROPOSED PHASE NOISE FILTER

The core technique in phase noise filter/cancellation is the frequency discriminator (FD). This is because the success of the whole system relies on the accurate phase noise extraction by FD. In addition, the FD determines the phase noise filter/cancellation sensitivity. So far, there are two approaches for the FD scheme.

Manuscript received December 2, 2016; revised March 9, 2017 and April 21, 2017; accepted April 27, 2017. Date of current version September 27, 2017. This work was supported by the DARPA ACT Program. This paper was recommended by Associate Editor Y. Sun. (*Corresponding author: Qun Jane Gu.*)

The authors are with the Department of Electrical and Computer Engineering, University of California at Davis, Davis, CA 95616 USA (e-mail: tnhu@ucdavis.edu; slhao@ucdavis.edu; jgu@ucdavis.edu).

Color versions of one or more of the figures in this paper are available online at <http://ieeexplore.ieee.org>.

Digital Object Identifier 10.1109/TCSI.2017.2703104

1549-8328 © 2017 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See http://www.ieee.org/publications_standards/publications/rights/index.html for more information.

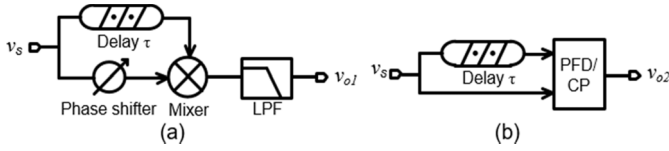


Fig. 1. Block diagram of (a) mixer based FD, (b) PFD/CP based FD.

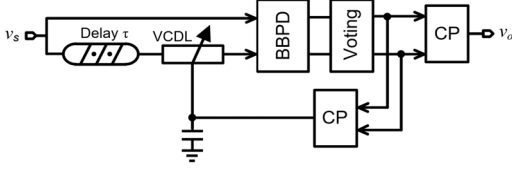


Fig. 2. Block diagram of the proposed BBPD based FD.

A. Mixer Based FD

Reference [10] has implemented the FD with a mixer. Fig. 1(a) shows the block diagram. Quadrature-phase inputs to the mixer are guaranteed with the help of a phase shifter. After the low pass filter (LPF), only the low frequency term remains at the output. The drawback is that the quadrature phase shifter requires complicated circuits. Furthermore, the output contains the amplitude information, which indicates that the mixer type FD is also sensitive to the amplitude noise.

B. Linear PFD/CP Based FD

The other type of FD is realized with a linear PFD/CP as the phase detector [7]. Its block diagram is shown in Fig. 1(b). If delay time τ is an integer number of the signal period, the delay line can convert the absolute input phase error into the relative phase difference. Then the linear PFD detects the relative phase difference and converts it to a voltage signal through the charge pump (CP). This approach enables in-phase inputs to the linear PFD, eliminating the need for a complex phase shifter. Moreover, the linear PFD only extracts the phase information, so it is insensitive to the amplitude noise. However, this FD scheme suffers from poor noise because the linear PFD with small gain ($1/2\pi$) amplifies the CP noise.

To overcome the disadvantages of the methods mentioned above, we employ a BBPD based FD, as shown in Fig. 2. It is composed of a long delay line, a voltage-controlled-delay-line (VCDL), a BBPD, a voting machine and a CP, and works in a similar way to the linear PFD/CP based FD. The difference is the use of a binary phase detector, which provides only an indication of the phase difference between the two inputs in a binary form. This inevitably pushes the phase detector gain to an extremely high level, thus suppressing the CP induced noise to a negligible value [11]. This approach also enables in-phase inputs, which is adjusted by the VCDL. We employ this BBPD based FD in the proposed phase noise filter (PNF) in Fig. 3.

This PNF consists of a voltage-controlled-phase-shifter (VCPS), a divider, a FD and a baseband circuit. The 10 GHz input LO is converted to a 1.25 GHz signal by the divide-by-8 divider to alleviate the design challenges of the subsequent circuits. The FD converts the phase noise information into a low frequency voltage and is further

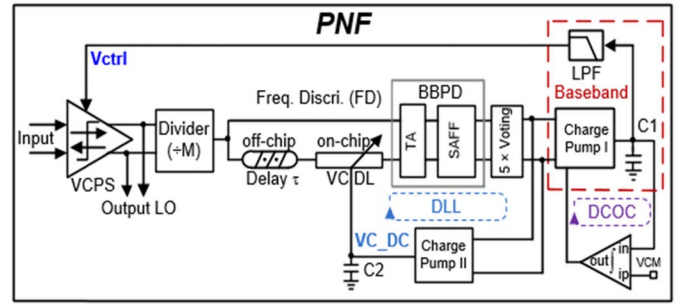


Fig. 3. Block diagram of the proposed BBPD based PNF.

filtered to get the desired control voltage (V_{ctrl}). The V_{ctrl} controls the VCPS in a negative feedback loop to achieve the required phase shifting in order to compensate the input phase error. As mentioned before, the FD is the key unit of this system because it determines the phase noise floor, suppression bandwidth and input frequency range.

The off-chip passive delay line is used to reach a long delay time τ with low noise, while the tunable on-chip VCDL with 250 ps tunable range is only employed to adjust the BBPD input phase relations. The adoption of a $5 \times$ voting scheme improves the BBPD noise performance by minimizing its error decisions, as will be discussed in the following section. Two local feedback loops are embedded into the PNF for calibration.

1) *Delay-Locked Loop (DLL)*: It is adopted to align the BBPD inputs. Typically, the delay line output is selected as the DLL output, which has the jitter peaking effect [24]. It occurs because DLL cannot distinguish between input clock jitter and output clock jitter. As mentioned [24], jitter peaking in the general DLLs can be reduced by decreasing the tracking bandwidth. Unlike the general DLLs, the output of this DLL is the phase difference in front of BBPD. It is worth noting that this DLL also requires a small tracking bandwidth to extract the phase difference. This means that any sudden jitter of input clock will appear at this DLL output before the loop reacts in the correct direction and reaches a new steady state. Therefore, this DLL is first-order with a high-pass response.

2) *DC Offset Cancellation (DCOC) Loop*: If there is circuit mismatch, the CP output will be saturated. To address this issue, the DCOC loop is applied to compensate the circuit mismatches. The integrator in the DCOC loop extracts the DC signal of the CP output, and then compares it with the common mode voltage (VCM). At last, the integrator stabilizes the CP output by adjusting the CP current. Since the proposed BBPD based PNF is based on a multiple-loop structure, the interaction among the loops should be considered. The main loop is made much faster than the DLL and DCOC loop to avoid interference. The DLL is used only to remove low frequency contents, including the DC component, thus having negligible effects to the main loop. With a 2 uA CP2 current and 80 pF C2, the DLL tracking bandwidth is close to 100 kHz. The same design strategy also applies to the DCOC loop. Its loop bandwidth needs to be significantly lower than the target phase noise suppression band to avoid interfering with the main loop. Consequently, DCOC only

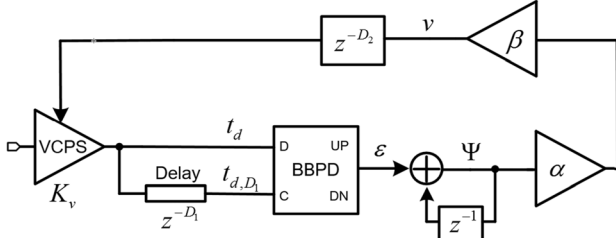


Fig. 4. Block diagram of the simplified discrete PNF model.

determine the DC level of the feedback voltage V_{ctrl} . Its simulated 3 dB bandwidth is about 10 Hz.

III. LIMIT CYCLE REGIME AND RANDOM NOISE REGIME ANALYSIS

The distinct feature of the BBPD based PNF is the nonlinearity brought in the loop, invalidating the traditional Laplace domain analysis used in linear systems. Analogous to any quantized systems, like DPLL, the fundamental aspect of BBPD based PNF is the presence of limit cycles when there is no random noise [12]. It is well known that the limit cycles will degrade the phase noise floor of the system. Therefore, we analyze the limit cycle regime of this PNF in the first step.

A. Limit Cycle Regime

For the sake of simplicity, we assume the PNF has no divider, no voting machine. Fig. 4 shows the block diagram of this simplified discrete PNF model. BBPD compares the rising edges of the D input (whose time instants is referred to as t_d) with those of the C input (whose time instants can be referred to as t_{d,D_1}). The binary output ε of BBPD is mapped to a numerical value +1 if the D input leads the C input, otherwise it is mapped to -1 [13]. We can write:

$$\varepsilon = \text{sgn}(\Delta t), \Delta t = t_d - t_{d,D_1} - D_1 \cdot T_o \quad (1)$$

where Δt shows the time difference between the BBPD two inputs, D_1 represents the number of clock delays, T_o indicates the clock period. By subtracting the $D_1 T_o$, time instants of the D rising edge t_d and the C rising edge t_{d,D_1} will arrive at the same time if assuming no jitter. Note that α represents the charge pump (CP) gain coefficient and Ψ indicates the integration character of the capacitor that follows the CP. The other blocks, such as the low pass filter (LPF), are modeled by βz^{-D_2} to express the gain and loop latency effects.

The voltage-controlled-phase-shifter (VCPS) is modeled as a linear block that provides the time shift of T_v depending on the feedback voltage v , which is given by:

$$T_v = -K_v \cdot v \quad (2)$$

where K_v is the time shift gain constant. In this model, we assume the response of the VCPS to a change of v is immediate without any transient behavior.

At the k -th clock cycle, the time occurrence t_d is adjusted by the time shift T_v in VCPS, according to the following:

$$t_d[k] = k \cdot T_o + T_v[k] \quad (3)$$

By combining the results so far achieved, the dynamics of the BBPD based PNF in Fig. 4 can be described by the following set of equations (k indicates the k -th clock cycle in the PNF model):

$$\begin{cases} \Delta t[k] = t_d[k] - t_{d,D_1} - D_1 \cdot T_o \\ \varepsilon[k] = \text{sgn}(\Delta t[k]) \\ \Psi[k] = \Psi[k-1] + \varepsilon[k] \\ v[k] = \alpha \beta \cdot \Psi[k] \\ T_v[k] = K_v \cdot v[k - D_2] \\ t_d[k+1] = t_d[k] - T_v[k] + T_v[k+1] + T_o \end{cases} \quad (4)$$

By adding the last equation in (4) for D_1 times, then substituting it in (4), the dynamics above can be simplified to:

$$\begin{cases} \Delta t[k + D_1] = -\alpha \beta K_v \cdot (\Psi[k + D_1 - D_2] - \Psi[k - D_2]) \\ \Psi[k] = \sum_{x=0}^k \text{sgn}(\Delta t[x]) \end{cases} \quad (5)$$

The definition of the following quantity:

$$\lambda \triangleq \frac{\Delta t}{\alpha \beta K_v} \quad (6)$$

allows the set of equations to be rewritten in the form:

$$\lambda[k + D_1] = - \sum_{x=k+1}^{k+D_1} \text{sgn}(\lambda[x - D_2]) \quad (7)$$

The evolution of the system in time is dominated by the above nonlinear map. $\alpha \beta K_v$ is the quantization step of the VCPS, λ is the time difference Δt normalized to this quantization step. In this system, we are interested in deriving the expression of the limit cycle period and the associated jitter. Indeed, once the operation trajectory of (7) is known, both properties can be evaluated. This will give insights into the nonlinear dynamics.

Eq. (7) is discrete, and is hard to solve for directly. However, an observation of (7) reveals that the trajectory of $\lambda[x]$ will move in integer steps in either the positive or negative direction. $\lambda[x]$ cannot stay positive or negative all the time since the term $-\text{sgn}(\lambda[x])$ will invert it. Hence, the operation trajectory must be periodic with a cycle period T_λ . However, the simultaneous presence of D_1 and D_2 in the map increases the difficulty of analyzing (7). As such, we decide to consider their effects separately by performing the analysis in three steps. In the first two steps, D_1 and D_2 will assume to be equal to one independently, which is already the minimum acceptable value for them in (7). Then the third step removes the previous assumption.

First, we just take the case of $D_1 = 1$, thus (7) simplifies to:

$$\lambda[k + 1] = -\text{sgn}(\lambda[k - D_2]) \quad (8)$$

If the trajectory starts with a positive value, with $(1+D_2)$ iterations, the trajectory will reach a negative value. After another $(1+D_2)$ iterates, the trajectory will become positive again. This means that the trajectory is periodic with a period of $2(1+D_2)T_o$.

Similarly, we also consider the case of $D_2 = 1$, and the nonlinear map (7) reduces to:

$$\lambda[k + D_1] = - \sum_{x=k}^{k-1+D_1} \text{sgn}(\lambda[x]) \quad (9)$$

In (9), there are D_1 iterates of $\text{sgn}(\lambda[x])$ for each λ value. The trajectory will eventually reach a steady state. At that time, $\lambda[x]$ must be half positive and half negative within the D_1 iterations. Otherwise, the trajectory will move in one direction, and indefinitely stay in the positive or negative axes. As such, it is fair to assume this trajectory has a period of about $D_1 T_o$.

Lastly, the above results can be combined together, so that the overall orbit period T_λ is affected by D_1 and D_2 simultaneously. Therefore, it is reasonable to predict that T_λ is the sum of two terms. One term $2(1+D_2-1)T_o$ depends only on the loop latency D_2 . The other term $(D_1-1)T_o$ depends on the clock delay D_1 . In both terms, D_1 and D_2 need to minus one independently in order to compensate the effect of being one in (8) and (9). So we get $T_\lambda = (2D_2 + D_1 - 1)T_o$.

To evaluate the validity of the predicted T_λ , we compute (7) with different D_1 and D_2 values based on two assumptions. First, the starting $(D_1 + D_2)$ iterations are meaningless, and $\lambda[x]$ will maintain the same value during this time. Therefore, we assume $\text{sgn}(\lambda[x]) = -1$ in the first $(D_1 + D_2)$ iterations. Second, $\lambda[x]$ may be equal to 0, so we suppose $\text{sgn}(0) = -1$. Computation results confirm that our prediction $T_\lambda = (2D_2 + D_1 - 1)T_o$ is correct.

In the ideal case, the absolute time jitter is $\Delta t_d[k] = t_d[k] - kT_o$. Obtaining the cycle period T_λ is already enough to calculate the peak-to-peak quantization induced jitter ($\Delta t_{d,pp}$) as the difference between the maximum and the minimum values that Δt_d assumes in the orbit. Indeed, Δt_d is a linear response to ε , which is equal to $\text{sgn}(\lambda)$. With a triangular shaped trajectory, a worst-case estimate for the $\Delta t_{d,pp}$, can be derived as:

$$\Delta t_{d,pp} = \frac{T_\lambda}{2} \cdot \alpha \beta K_v = \frac{(2D_2 + D_1 - 1)T_o}{2} \cdot \alpha \beta K_v \quad (10)$$

In (10), $\Delta t_{d,pp}$ is already denormalized by multiplying $\alpha \beta K_v$ on the right hand. This is to evaluate the effects of PNF parameter variations on $\Delta t_{d,pp}$. The standard deviation of the quantization jitter is given by [14]:

$$J_q = \frac{\Delta t_{d,pp}}{\sqrt{12}} = \frac{(2D_2 + D_1 - 1)T_o}{4\sqrt{3}} \cdot \alpha \beta K_v \quad (11)$$

Eq. (11) reveals that the larger VCPS phase granularity ($\alpha \beta K_v$) leads to the larger quantization jitter J_q . Additionally, (11) suggests that quantization jitter increases as the clock delays D_1 , a parameter which cannot be reduced below a certain value for PNF noise sensitivity.

To further illustrate the dynamics of the PNF in limit cycle regime, a frequency modulated (FM) single tone (ST) input is applied to simulate the ideal PNF model, which adopts the delay time $D_1 T_o$ of about 20 ns. The FM signal is at 1 MHz offset frequency. The simulated CP output is plotted in Fig. 5, and presents that BBPD is flipping with a period of about 41 ns.

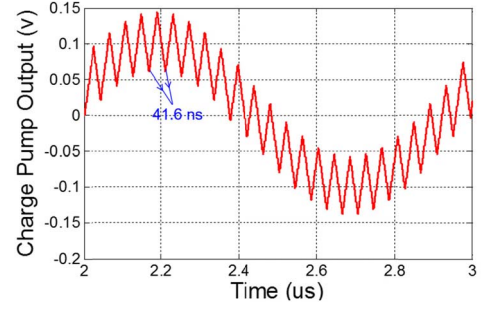


Fig. 5. Simulated waveform of the CP output in the ideal PNF model with 1 MHz FM ST input.

B. Random Noise Regime

We have so far analyzed the BBPD based PNF neglecting the presence of any random noise sources. In practice, when the circuit noise is large enough, it acts as a dithering signal for the BBPD, which can be then linearized. This operation mode is referred to as the random noise regime [12]. The derivation of a linearized model of the BBPD has extensively been discussed in literature. For input signals around its threshold, the BBPD behaves as a linear block given that its averaged output is proportional to its input with a gain of [2]:

$$K_{BBPD} \approx \frac{0.8T_{ref}}{2\pi\sigma_t} \quad (12)$$

Comparing with the gain of $(1/2\pi)$ in the conventional three-state PFD, the BBPD has a much larger gain. If assuming the RMS value of input jitter σ_t in the range of 0.5 ps or less, the BBPD gain is four orders of magnitude higher than the conventional PFD gain.

The other blocks in the loop are ideally linear and thus they can be described in the frequency domain. Now, the Laplace analysis can be adopted to get the PNF phase noise transfer function:

$$H_{PNF} = \frac{1}{1 + K_{VCPS} \cdot H_{DLL} \cdot K_{BBPD} \cdot H_{BB}/M} \quad (13)$$

where K_{VCPS} is the gain of VCPS; M is the division ratio; H_{DLL} shows the transfer function of the delay-locked-loop (DLL), while H_{BB} represents the transfer function of the baseband circuit blocks including the CP1, C1 and LPF. The CP1 current is adjustable to achieve different gain modes for H_{PNF} . The input and output signals of (13) are $\varphi_{n,in}$ and $\varphi_{n,out}$ respectively, which have been marked in Fig. 7. As mentioned earlier, H_{DLL} has a high pass response, calculated as [23]:

$$H_{DLL} = \tau s \cdot \frac{s/\omega_p}{1 + s/\omega_p} \quad (14)$$

In (14), ω_p is a low frequency pole determined by the DLL tracking bandwidth, τ is the delay time. The input and output signals of (14) are $\varphi_{nD,IV,out}$ and $\varphi_{nD,LL,out}$ respectively. Together with the low pass filtering feature of the PNF main loop, the overall PNF open loop demonstrates a band-pass characteristic. Hence, the PNF close loop is able to suppress the input phase error within the band, as shown in Fig. 6. The lower frequency boundary is determined by the DLL while the

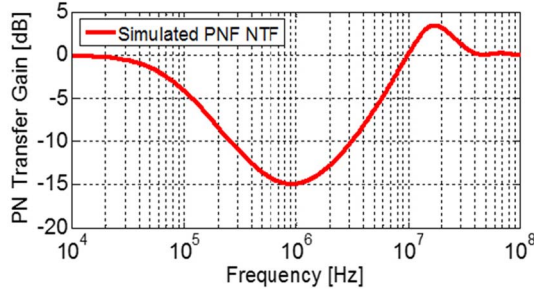


Fig. 6. Simulated PNF phase noise transfer function.

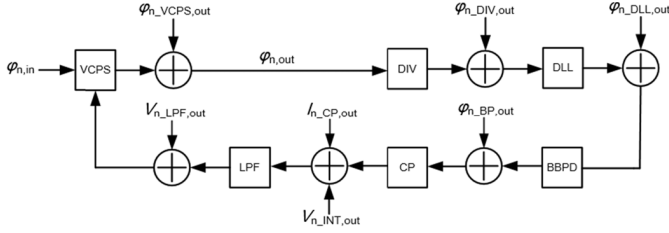


Fig. 7. Linear noise model for PNF.

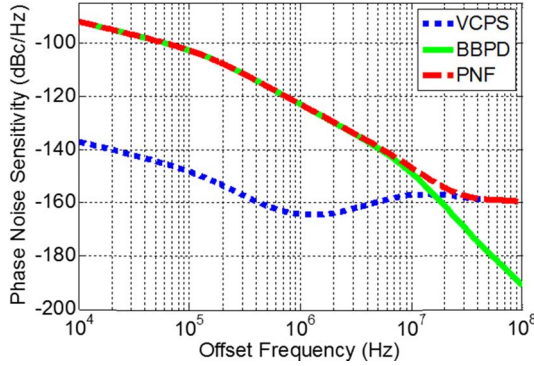


Fig. 8. PNF phase noise floor and major noise contributions.

higher frequency corner comes from the LPF. A wider suppression band requires a higher corner frequency of the LPF. However, this may lead to a limited phase margin in the open loop. So the LPF must be designed carefully to guarantee both the suppression bandwidth and loop stability. The same method can now be applied to calculate the phase noise floor in the PNF with a linear noise model, as shown in Fig. 7. It is one of the most important specifications, because it determines the best phase noise the PNF can achieve. Based on the derived noise transfer functions and the simulated output noise of each block, the PNF phase noise floor and major noise contributions are shown in Fig. 8. It clearly presents that the BBPD contributes most of the noise. This makes sense because all noise sources after the BBPD can be minimized to a negligible level due to its large gain. Besides, through simulation, the VCPS, divider and delay line they all have very good phase noise. As a consequence, BBPD itself becomes the dominant noise source. The BBPD noise induced jitter is:

$$J_r \approx \frac{T_{ref}}{2\pi} \cdot \sqrt{2 \int_0^\infty \frac{S_{BBPD}(f)}{|H_{DLL}(f)/M|^2} df} \quad (15)$$

where S_{BBPD} represents the BBPD input noise power spectral density (PSD). Fig. 8 leads us to the conclusion that a worthy effort in the PNF design is to minimize the BBPD noise so as to enhance the PNF noise sensitivity. This will be discussed in the next section.

In reality, the total output jitter J_o follows the sum of the two noise contributions, the quantization induced jitter and the random noise induced jitter [12], and can be expressed below under the premise that $\tau = D_1 T_o$:

$$J_o \approx \frac{T_{ref}}{2\pi} \cdot \sqrt{2 \int_0^\infty \frac{S_{BBPD}(f)}{|2\pi f D_1 T_o / M|^2} df} + \frac{(2D_2 + D_1 - 1)T_o \cdot \alpha\beta K_v}{4\sqrt{3}} \quad (16)$$

Eq. (16) provides a few key observations. While the quantization induced jitter J_q grows as the number of clock delays D_1 , the noise induced jitter J_r decreases as D_1 . As such, the total jitter given by (16) can exhibit a minimum value. The optimum D_1 is found when the two terms on the right hand side of (16) are equal, or in other words when the jitter induced by quantization equals the jitter induced by random noise. In addition, a compromise must be made between the noise sensitivity and noise suppression, as noise sensitivity prefers a small VCPS phase granularity $\alpha\beta K_v$ while noise suppression prefers a large one. Besides, the loop latency D_2 should be kept small in order to minimize the quantization jitter.

IV. PROPOSED PHASE NOISE FILTER IMPLEMENTATION

A. Voting Machine

In last section, we already demonstrate that in the ideal case with no random noise, BBPD will flip with a constant cycle period so that the charge pump (CP) output has a saw-tooth shaped waveform. However, with the presence of random noise at the BBPD input, the saw-tooth waveform may be corrupted since undetermined decisions may appear at the BBPD output.

The working scheme of a BBPD is similar to that of a comparator. Consider a comparator with a certain input referred noise, denoted by its standard deviation σ_{input} . Due to this noise, the comparator has a certain output probability function, which logically means that the comparator makes error decisions when the input signal is close to σ_{input} . Reference [15] indicates that when the same comparator is used several times to repeat the same decision, a majority voting logic can be applied on the obtained output samples to decide the final output. Using more samples increases the possibility that the majority decision is correct. In view of this, majority voting helps to scale down the comparator noise. Obviously, this approach can also be adopted to reduce the BBPD noise. However, it is hard to directly apply the technique in [15] to our PNF by using the BBPD several times to repeat the same decision, because the BBPD input is not a voltage signal, but a time difference signal. Currently, it is still challenging and power consuming to hold a time difference signal in circuit.

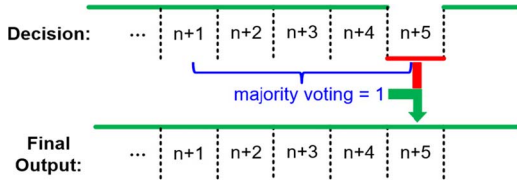
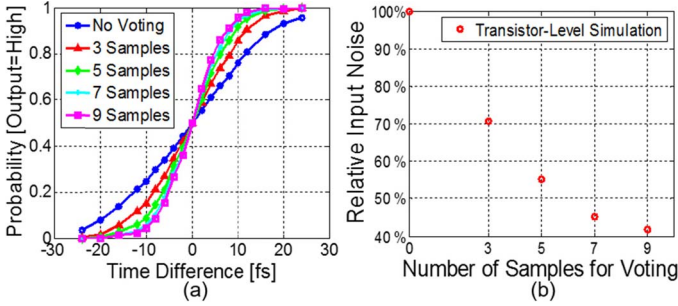
Fig. 9. Work diagram for a $5 \times$ voting scheme.

Fig. 10. (a) Probability of BBPD output as high vs. input in different voting schemes and (b) relative BBPD equivalent input referred noise vs. number of samples in the voting scheme.

Fortunately, the unique feature of our PNF allows the employment of majority voting by holding previous BBPD decisions. In the ideal case, BBPD will maintain the same decision for the half cycle period, which is about 20 ns. Given the 1.25 GHz input frequency, BBPD can hold the same decision for at least 25 consecutive periods. This feature implies that one could minimize error by performing majority voting for several consecutive decisions. To better illustrate this working mechanism, a $5 \times$ voting scheme is taken as an example and is shown in Fig. 9. The $(n+1)$ -th to $(n+4)$ -th decisions are all one (green signal), and the error bit zero (red signal) at the $(n+5)$ -th decision can be corrected by majority voting for the recent 5 decisions. The amount of noise reduction can be varied by applying different number of samples used to determine the majority decision. To intuitively present the effect on equivalent noise for different number of samples, Fig. 10(a) illustrates the transfer curves of a transistor-level BBPD with different samples. This figure also indicates the statistical results of the equivalent noise in different voting scheme, as shown in Fig. 10(b). It presents that voting on 5 samples reduces the effective noise to about 55% of the original value. With more samples, the equivalent noise improvement slope is reduced. In addition, more samples also cause larger loop latency. As mentioned previously in Section III B, large loop latency D_2 degrades the quantization induced jitter. Finally, we choose to use the $5 \times$ voting scheme as the optimum option, as shown in Fig. 11. Four registers are employed to hold the previous 4 BBPD decisions, together with the latest one. All 5 samples are fed into the majority voting logic to decide the final output.

If we deliberately add some small noise to the PNF model, the original saw-tooth like CP output will be corrupted. The top waveform in Fig. 12 plots the simulated CP output in the noisy model without voting, while the bottom waveform in Fig. 12 plots the CP output with the $5 \times$ voting scheme. It is clear that the corrupted saw-tooth waveform is partly recovered

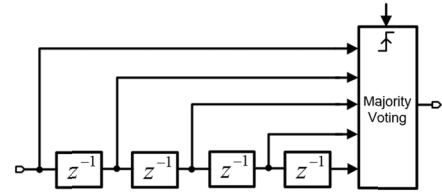
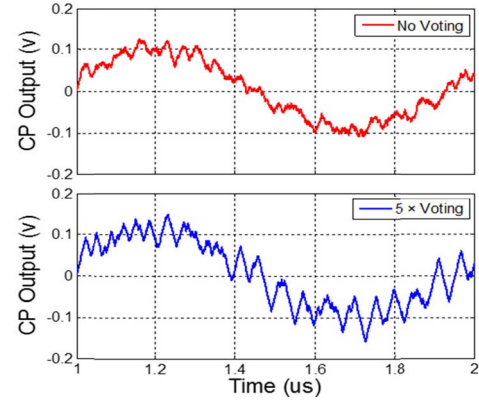
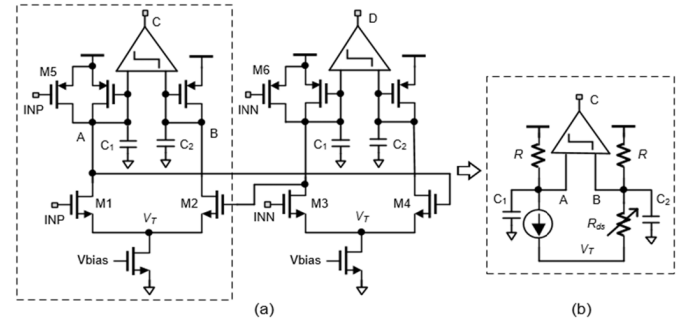
Fig. 11. Block diagram of the proposed $5 \times$ voting machine.Fig. 12. Simulated CP output in the noisy PNF model without and with the $5 \times$ voting scheme.

Fig. 13. (a) Schematic of the TA and (b) equivalent model for the left side of the TA.

by the majority voting because of the reduced equivalent BBPD noise.

B. TA In BBPD

In order to further reduce the BBPD noise, the BBPD itself also requires careful design. Researchers typically adopt a D-flip-flop (DFF) as BBPD. However, in high frequency applications, DFF may experience metastability and hysteresis effects, thereby generating undetermined outputs [16], and increasing the BBPD noise. So, to address this issue, a time-amplifier (TA) is included in the BBPD to enlarge the time difference to make the DFF work away from these regions. Fig. 13(a) shows the schematic of the TA [17]. Over the years, many techniques have been published about TA with the focus on linearity [18], [19], because TA is often employed in TDC to enhance its resolution by amplifying the time residue. Its accurate amplification gain is a key factor in this application. However, in our PNF, the design consideration for the TA

was optimizing its noise performance, since extra noise in the BBPD is undesirable. In addition, there is no strict requirement for TA in terms of linearity since the proposed PNF is already a nonlinear system. In other words, our proposed design sacrifices the other performances, such as gain and linearity, to achieve a low noise TA for the PNF.

The transient behavior of this TA is that upon the arrival of a rising edge at M1, a current impulse travels from the drain of M1 to the drain of M2. The comparator detects its input phase inversion [17]. To reset the TA every period, M5-M6 are employed. The circuit operation for the left side of the TA can be approximated by the model depicted in Fig. 13(b). Here, M1 is modeled as a current source, whereas M2 is modeled as a linear resistance R_{ds} . Also, the biasing current source is neglected since it is assumed that the traveled current impulse is larger than the tail current. For the sake of simplicity, our design applies the method in [20] to analyze the phase noise of node C in order to guide the low noise TA design.

The next simplifying assumption is that M1 operates in a similar way to the NMOS transistor in an inverter on the falling edge at the output. Moreover, as illustrated in Fig. 13(b), the diode-connected PMOS transistor acts as a resistor, producing noise across C_1 . Hence, according to [20], the phase noise on the falling edge at node A is equal to:

$$S_A \approx \left\{ \frac{\pi^2}{s_r^2 C_1^2} [\Delta T f_{in} S_I(f) + \Delta T^2 f_{in}^2 S_{1/f}(f)] \right\} + \frac{\pi^2 f_{in}^2 \cdot 4kTR}{s_r^2} \quad (17)$$

where the first item on the right evaluates the M1 contributed phase noise, and the second item denotes the resistor induced phase noise from the diode-connected PMOS. Note that s_r presents the slew rate, ΔT denotes the duration time for the node transition, C_1 is the output load capacitance, $S_I(f)$ shows the active device thermal noise, $S_{1/f}(f)$ indicates the active device flicker noise and R is resistance of the diode-connected PMOS.

Similarly, the phase noise on the rising edge at node B is given by [20]:

$$S_B \approx \left\{ \frac{\pi^2}{s_r^2 C_2^2} [\Delta T f_{in} S_I(f) + \Delta T^2 f_{in}^2 S_{1/f}(f)] \right\} + \frac{\pi^2 f_{in}^2}{s_r^2} \cdot \frac{kT}{C_2 f_{in}} + \frac{\pi^2 f_{in}^2 \cdot 4kTR}{s_r^2} \quad (18)$$

where the second item on the right hand denotes the kT/C_2 noise from the resistor R_{ds} [20], and C_2 represents the load capacitance.

With the analysis developed above, we now seek to optimize the TA design for low noise. As can be seen from the ΔT in (17) and (18), the phase noise rises in proportion to the transition time ΔT . Therefore, the channel length of NMOS transistors must be the smallest since the longer channel inevitably raises transition time ΔT . Then we make several approximations: (a) device thermal noise spectrum $S_I(f) = 4kT\gamma g_m$, where $g_m = 2I_D/(V_{GS}-V_{th})$, I_D is the drain current and we have $I_D \propto W$ at the point of interest,

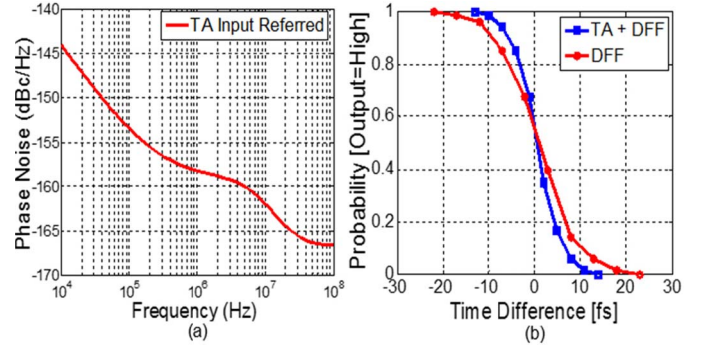


Fig. 14. (a) Simulated input referred phase noise of TA and (b) probability of TA + DFF and DFF outputs as high vs. input time difference.

where W represents the channel width of the transistor. So we get $S_I(f) \propto W$. Similarly, $S_{1/f}(f) = g_m^2 K/(C_{ox}WLf) \propto W$. (b) The slew rate $s_r \propto I_D/C_L \propto W/C_L$, where C_L represents the load capacitance. (c) The transition time window width, ΔT , is proportional to $1/s_r \approx C_L/W$ [20].

If assuming an ideal comparator, the overall phase noise of node C is proportional to the phase noise in node A and B, which are now rewritten as:

$$S_A(f) \propto \frac{C_1(\eta W + C_1)}{W^3}, S_B(f) \propto \frac{C_2(\eta W + C_2)}{W^3} \quad (19)$$

An important point emerging from the above derivation is that low noise design of TA favors wide channel width NMOS transistors and small load capacitance. On the other hand, the PMOS and tail current have little effect on the TA noise. They only affect the DC operating point. However, enlarging the NMOS channel widths too much is not favorable, not only for the reason of power consumption but also due to the loading effect. Large load capacitance caused by the NMOS transistors would inevitably degrade the buffer phase noise.

To simulate the TA input referred phase noise, we embed the TA with a PFD/CP to run a Pnoise analysis, and deduce the phase noise from the CP output noise, as shown in Fig. 14(a). To further illustrate the DFF noise reduction by the TA, transient noise simulations are also performed to compare the transition curves of TA + DFF and DFF, as illustrated in Fig. 14(b). It is clear that the application of TA shrinks the DFF transition region because of the reduced equivalent noise.

C. SAFF In BBPD

A sense-amplifier-flip-flop (SAFF) is chosen to realize the DFF in BBPD because it exhibits a good metastability among DFFs [21]. However, the conventional SAFF also suffers from hysteresis effect because any master-slave registers have a tendency to exhibit memory effect in the sampling process, thus negatively influencing the jitter performance [22]. As for the conventional SAFF, as shown in Fig. 15(a), there is kickback effect from the SR latch in the slave stage to the master stage, leading to bad hysteresis. Therefore, to reduce the hysteresis of the conventional SAFF, modification can be made by inserting isolation between the two stages, as shown in Fig. 15(b). The master stage is still a sense amplifier, but the slave stage changes from a SR latch to a cross coupled

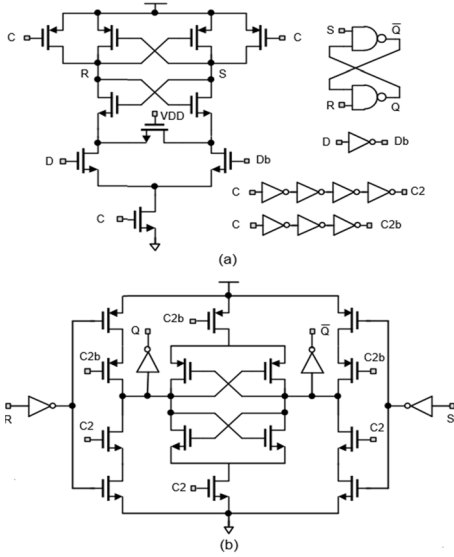


Fig. 15. (a) Schematic of the conventional SAFF and (b) modified slave stage in the SAFF.

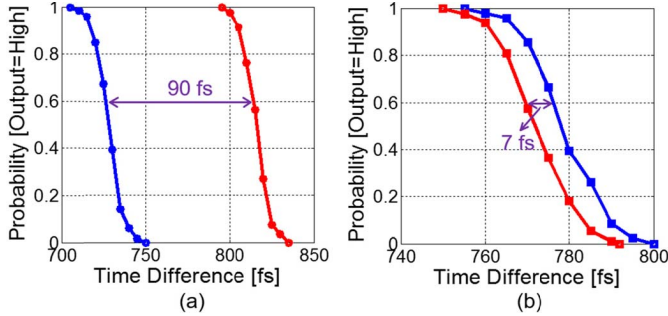


Fig. 16. Transient noise simulation results showing the hysteresis of the (a) conventional SAFF and (b) modified SAFF.

latch with an isolated stage. Clock for the slave stage is also delayed compared with the master stage.

To evaluate the hysteresis of the above SAFFs, transient noise simulations are also performed. The results are shown in Fig. 16. It exhibits the conventional SAFF has hysteresis of about 90 fs, but hysteresis of the modified SAFF drops to 7 fs.

D. Voltage Controlled Phase Shifter

Voltage-controlled-phase-shifter (VCPS) is a key block in PNF. The conventional passive phase shifters are not applicable in this design, because of the large insertion loss. For this reason, an active type VCPS with a cross-coupled pair is implemented [8], as shown in Fig. 17. The phase of output voltage is shifted by adjusting the resonant frequency of the tank. The resonant tank is composed by an inductor, a 5-bit capacitor array and a varactor.

This resonant tank can be represented by a parallel of an inductor L , a capacitor C and a resistor R . Its phase shift is realized by changing the capacitor value of the varactor or cap array. If the tank capacitor changed by $\pm\Delta C$, the tank impedance phase shift is calculated as

$$\Delta\phi(Z_{in}) = \mp R \cdot \Delta C / \sqrt{LC} \quad (20)$$

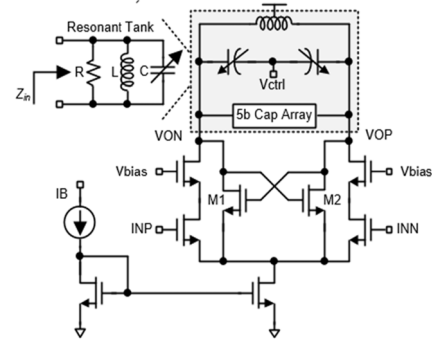


Fig. 17. Schematic of the VCPS.

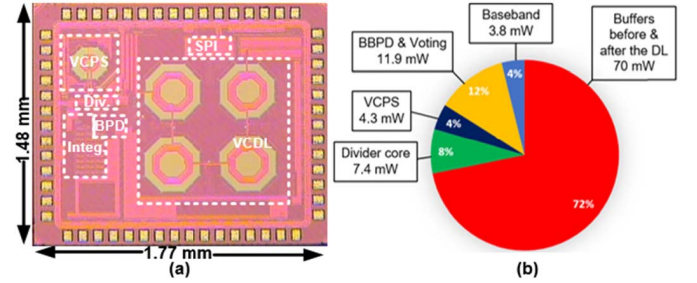


Fig. 18. (a) Die photo and (b) power breakdown.

As can be seen from (20), the phase shift is proportional to the capacitor change. The cross-coupled pair (M1 and M2) is used as a Q booster to enhance the parallel tank resistance to generate large output.

V. MEASUREMENT RESULTS

The proposed BBPD based PNF was fabricated in a 65 nm CMOS technology and the chip area is 1.77 mm $\times$ 1.48 mm. Fig. 18(a) shows the die photo. A 20 ns off-chip delay line is adopted in the measurement. We have used two methods to test the noise suppression and noise sensitivity based on two input signals, one is the frequency modulated (FM) single tone (ST) input, the other one is the wideband noise input. Both PNF input signals are generated by the signal generator E8257D. After that, signal analyzer measures the PNF output spectrum. The feedback voltage (V_{ctrl}) to the VCPS is monitored by the oscilloscope. Fig. 18(b) shows the power breakdown. Buffers before and after the DL consume 72% total power to drive the delay line.

A. FM Input Measurement

A FM ST signal is applied to emulate the LO with phase noise at a given offset frequency. We first measure FM signal at 1 MHz offset. Fig. 19 shows the 10.06 GHz FM input (red curve) and output (blue curve) spectra of PNF. It presents that the small and large phase noise suppression level is 15/23 dB respectively for the 10.06 GHz carrier at 1 MHz offset.

The input signal is generated by the signal generator, which is of extremely good noise floor. The uplifted noise floor at PNF output is due to its own circuit noise. This illustrates that PNF's noise floor determines the best sensitivity it can achieve.

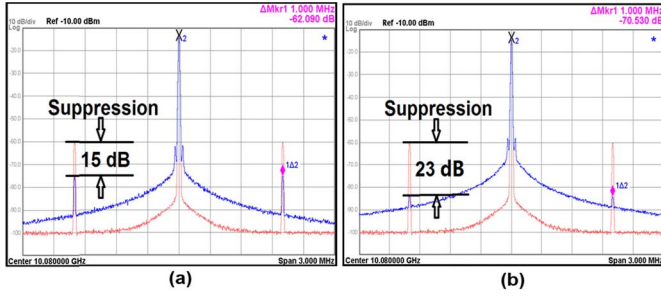


Fig. 19. Input and output spectra of the PNF with the FM ST input with (a) small suppression and (b) large suppression.

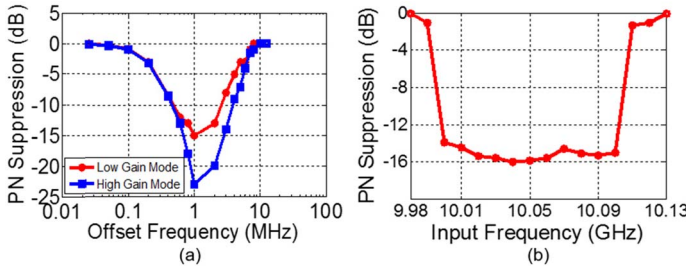


Fig. 20. (a) Phase noise suppression vs. offset frequency in low/high gain mode and (b) phase noise suppression vs. input frequency.

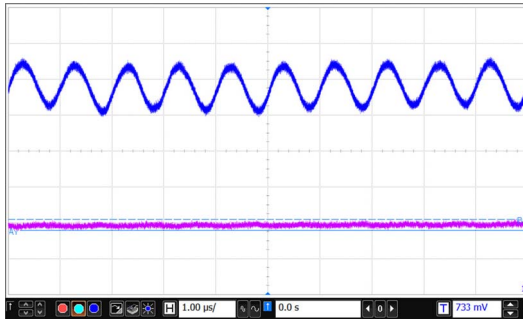


Fig. 21. Measured V_{ctrl} and VC_{DC} waveform.

If the input phase noise is already better than PNF's noise floor, there will be no noise suppression. In this case, input phase noise is only suppressed at 1 MHz offset frequency.

By sweeping the FM rate, we get the noise suppression bandwidth. Fig. 20(a) plots the noise suppression level versus offset frequency at the two different gain modes. The suppression offset frequency is from 100 kHz to 8 MHz and the suppression level is 15/23 dB in the low/high gain mode at around 1 MHz offset, respectively. Fig. 20(b) shows the noise suppression versus the input frequency at 1 MHz offset in low gain mode. Within the input frequency range of 10-10.1 GHz, the PNF has similar noise suppression level. Once out of the range, PNF will have no suppression for the reason that the delay-locked-loop (DLL) cannot lock anymore. This is limited by the tunability of VCDL. The blue waveform (V_{ctrl}) shown in Fig. 21 indicates the 1 MHz FM information has been successfully extracted and feedback to VCPS. The purple waveform (VC_{DC}) in Fig. 21 is almost constant, demonstrating the slow response of DLL.

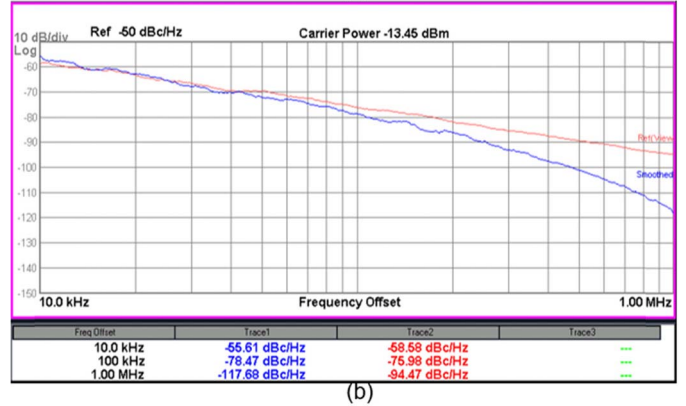
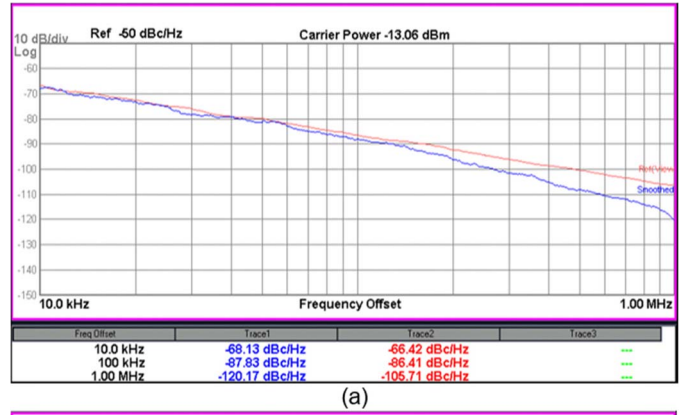


Fig. 22. Input and output spectra of the PNF with the wideband noise input in (a) low gain mode and (b) high gain mode.

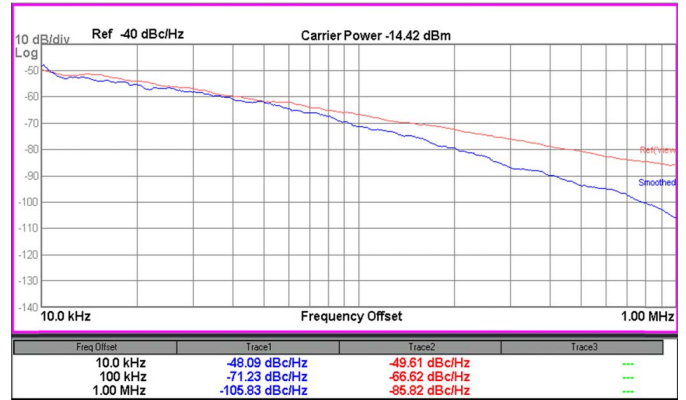


Fig. 23. Input and output spectra of the PNF with a large wideband noise input in high gain mode.

B. Wideband Noise Input Measurement

To further evaluate the PNF performance, the wideband noise input is also generated by the signal generator E8257D to test the noise suppression and sensitivity for the PNF. E8257D can only generate the wideband noise up to 1 MHz, which limits the highest offset frequency in the measurement.

Fig. 22 shows the input and output phase noise curves in the low/high gain modes. Red curve represents the input phase noise while blue curve represents the output phase noise. In the low gain mode, PNF can suppress the input phase noise down to -120.2 dBc/Hz at 1 MHz offset with 14.5 dB noise suppression. In the high gain mode, PNF has a

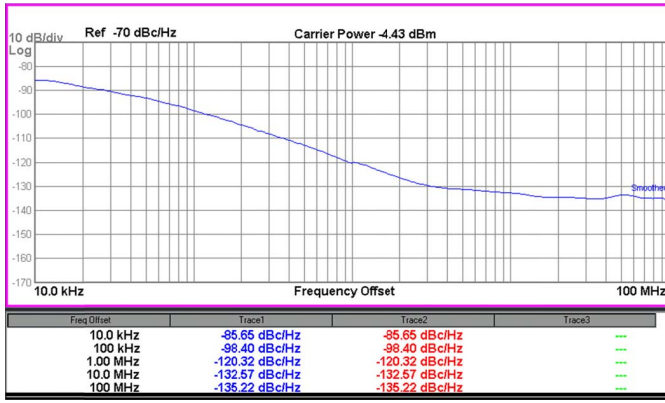


Fig. 24. Phase noise floor of the PNF in low gain mode.

TABLE I
PERFORMANCE SUMMARY AND COMPARISON WITH SOAs

	[6] TMTT 2015	[5] RFIC 2012	[8] IMS 2016	This work
Center Frequency	1.5 GHz	5 GHz	10 GHz	10 GHz
Delay Line	Off-chip high Q FBAR	On-chip active inverter	On-chip delay line+ off-chip SAW filter	On-chip delay line+ off-chip SAW filter
Phase Detector	Mixer	Mixer	PDF	BBPD
Input Bandwidth	< 5 MHz ^a	3.1-7.5 GHz	117 MHz	100 MHz
Need Reference Clock	no	yes	no	no
PN @ 1 MHz (Convert to 10 GHz)	-124 dBc/Hz	-99 dBc/Hz	-116 dBc/Hz	-120/-118 dBc/Hz ^b
PN Suppression @ 1 MHz	n/a	12.5 dB	10.6 dB	15/23 dB ^b
Power Consumption	350 mW	29.6 mW	102 mW	98 mW
Technology	0.13 μ m	90 nm	65 nm	65 nm

a. estimated from paper b. with low/high gain mode

larger suppression of 23.2 dB at 1 MHz offset, but can only suppress the input down to -117.7 dBc/Hz. Although a larger CP current is adopted in the high gain mode to improve the noise suppression, this is at the cost of a worse noise floor. As can be seen from (13), a larger CP current in the main loop contributes to a smaller $|H_{PNF}|$. On the other hand, as illustrated in (16), larger CP gain coefficient α will also degrade the total output jitter due to the enlarged quantization induced jitter. It is known that the gain of BBPD may vary as the input phase noise profile varies. Hence, a measurement result with a large input noise in high mode is also given in Fig. 23. This suppression is 3 dB smaller than that of Fig. 22(b) due to the reduced BBPD gain.

As mentioned, if injecting a clean carrier signal to the PNF, we can get the PNF noise floor from its output. Fig. 24 illustrates the noise floor in low gain mode. The measured phase noise floor is -120 dBc/Hz at 1 MHz offset. This also explains why we cannot achieve better output phase noise in Fig. 22. The integrated jitter in the suppression band (100 kHz – 10 MHz) is 81 fs.

Table I summarizes the performance of the proposed PNF and compares with state-of-the-arts (SOAs). Even though the -120 dBc/Hz at 1 MHz phase noise floor is a little worse than [6], the 100 MHz input bandwidth is much wider than that work. This work also exhibits the largest suppression among all works. Therefore, this PNF can be used as an individual block to suppress the phase noise of the LOs for performance boosting.

VI. CONCLUSION

A BBPD frequency discriminator (FD) based phase noise filter (PNF) is presented in this paper. The discriminator utilizes a BBPD to provide a large gain to improve PNF sensitivity by suppressing the charge pump (CP) noise. The best phase noise sensitivity is -120 dBc/Hz at 1 MHz offset and the maximum phase noise suppression is 23 dB. This PNF scheme also exhibits wide suppression bandwidth and wide input range. It can be utilized individually or embedded with clock generations to improve phase noise in a variety of systems.

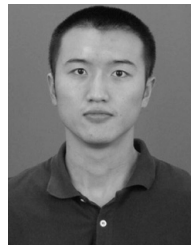
ACKNOWLEDGMENT

The authors would like to thank Dr. William Chappell and Dr. Troy Olsson for funding support.

REFERENCES

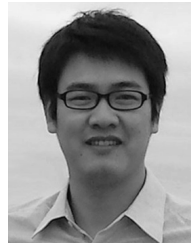
- [1] X. Gao, E. A. M. Klumperink, M. Bohsali, and B. Nauta, "A low noise sub-sampling PLL in which divider noise is eliminated and PD/CP noise is not multiplied by," *IEEE J. Solid-State Circuits*, vol. 44, no. 12, pp. 3253–3263, Dec. 2009.
- [2] S. Levantino, G. Marzin, C. Samori, and A. L. Lacaita, "A wideband fractional-N PLL with suppressed charge-pump noise and automatic loop filter calibration," *IEEE J. Solid-State Circuits*, vol. 48, no. 10, pp. 2419–2429, Oct. 2013.
- [3] H.-Y. Chang, Y.-L. Yeh, Y.-C. Liu, M.-H. Li, and K. Chen, "A low-jitter low-phase-noise 10-GHz sub-harmonically injection-locked PLL with self-aligned DLL in 65-nm CMOS technology," *IEEE Trans. Microw. Theory Techn.*, vol. 62, no. 3, pp. 543–555, Mar. 2014.
- [4] J. M. Ávila-Ruiz, I. Molina-Fernández, L. Moreno-Pozas, and A. Moscoso-Mártir, "Ultra-wideband VCO phase-noise reduction loop based on a six-port phase-shifterless delay-line discriminator," *IEEE Trans. Microw. Theory Techn.*, vol. 62, no. 12, pp. 3076–3084, Dec. 2014.
- [5] S. Min, T. Copani, S. Kiaei, and B. Bakaloglu, "A 90 nm CMOS 5 GHz ring oscillator PLL with delay-discriminator based active phase noise cancellation," in *Proc. IEEE Radio Freq. Integr. Circuits Symp.*, Jun. 2012, pp. 173–176.
- [6] A. Imani and H. Hashemi, "An FBAR/CMOS frequency/phase discriminator and phase noise reduction system," *IEEE Trans. Microw. Theory Techn.*, vol. 63, no. 5, pp. 1658–1665, May 2015.
- [7] S. Hao, T. Hu, and Q. J. Gu, "A 10 GHz delay line frequency discriminator and PD/CP based CMOS phase noise measurement circuit with -138.6 dBc/Hz sensitivity at 1 MHz offset," in *Proc. IEEE Radio Freq. Integr. Circuits Symp. (RFIC)*, Sep. 2015, pp. 63–66.
- [8] S. Hao and Q. J. Gu, "A 10 GHz phase noise filter with 10.6 dB phase noise suppression and -116 dBc/Hz sensitivity at 1 MHz offset," in *Proc. IEEE IMS*, May 2016, pp. 1–4.
- [9] S. Hao, T. Hu, and Q. J. Gu, "Phase noise improvement for array systems," in *IEEE MTT-S Int. Microw. Symp. Dig.*, May 2016, pp. 1–4.
- [10] W. Khalil, B. Bakaloglu, and S. Kiaei, "A self-calibrated on-chip phase-noise measurement circuit with -75 dBc single-tone sensitivity at 100 kHz offset," *IEEE J. Solid-State Circuits*, vol. 42, no. 12, pp. 2758–2765, Dec. 2007.
- [11] T. Hu, S. Hao, and Q. J. Gu, "A 10.01 - 10.1 GHz bang-bang PD based phase noise filter with 12.6 dB noise suppression," in *Proc. IEEE Int. Symp. Radio-Freq. Integr. Technol. (RFIT)*, Sep. 2016, pp. 1–3.
- [12] M. Zanuso, D. Tasca, S. Levantino, A. Donadel, C. Samori, and A. L. Lacaita, "Noise analysis and minimization in bang-bang digital PLLs," *IEEE Trans. Circuits Syst. II, Exp. Briefs*, vol. 56, no. 11, pp. 835–839, Nov. 2009.
- [13] N. D. Dalt, "A design-oriented study of the nonlinear dynamics of digital bang-bang PLLs," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 52, no. 1, pp. 21–31, Jan. 2005.
- [14] G. Marucci, S. Levantino, P. Maffezzoni, and C. Samori, "Analysis and design of low-jitter digital bang-bang phase-locked loops," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 61, no. 1, pp. 26–36, Jan. 2014.

- [15] P. Harpe, E. Cantatore, and A. van Roermund, "A 10 b/12b 40 kS/s SAR ADC with data-driven noise reduction achieving up to 10.1 b ENOB at 2.2 fJ/conversion-step," *IEEE J. Solid-State Circuits*, vol. 48, no. 12, pp. 3011–3018, Dec. 2013.
- [16] D. Rennie *et al.*, "Performance, metastability, and soft-error robustness trade-offs for flip-flops in 40 nm CMOS," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 59, no. 8, pp. 1626–1634, Aug. 2012.
- [17] M. Oulmane and G. W. Roberts, "CMOS digital time amplifiers for high resolution timing measurement," *Analog Integr. Circuits Signal Process.*, vol. 43, no. 3, pp. 269–280, Jun. 2005.
- [18] M. Lee and A. A. Abidi, "A 9 b, 1.25 ps resolution coarse-fine time-to-digital converter in 90 nm CMOS that amplifies a time residue," *IEEE J. Solid-State Circuits*, vol. 43, no. 4, pp. 769–777, Apr. 2008.
- [19] S.-K. Lee, Y.-H. Seo, H.-J. Park, and J.-Y. Sim, "A 1 GHz ADPLL with a 1.25 ps minimum-resolution sub-exponent TDC in 0.18 μ m CMOS," *IEEE J. Solid-State Circuits*, vol. 45, no. 12, pp. 2874–2881, Dec. 2010.
- [20] A. Homayoun and B. Razavi, "Analysis of phase noise in phase/frequency detectors," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 60, no. 3, pp. 529–539, Mar. 2013.
- [21] D. Li, P. Chuang, and M. Sachdev, "Comparative analysis and study of metastability on high-performance flip-flops," in *Proc. 11th Int. Symp. Quality Electron. Design (ISQED)*, Mar. 2010, pp. 853–860.
- [22] N. Da Dalt and T. G. Noll, *Theory and Implementation of Digital Bang-Bang Frequency Synthesizers for High Speed Serial Data Communications*. Aachen, Germany: RWTH Aachen Univ., 2007.
- [23] S. Hao, T. Hu, and Q. J. Gu, "A 10-GHz delay line frequency discriminator and PD/CP-based CMOS phase noise measurement circuit," *IEEE Trans. Microw. Theory Techn.*, vol. 65, no. 7, pp. 2361–2372, Feb. 2017.
- [24] M.-J. E. Lee and J. W. Dally, "Jitter transfer characteristics of delay-locked loops—Theories and design techniques," *IEEE J. Solid-State Circuits*, vol. 38, no. 4, pp. 614–621, Apr. 2003.



Tongning Hu (S'16) received the B.S. degree in optoelectronic engineering from the Nanjing University of Science and Technology, Nanjing, China, in 2010, and the M.S. degree in electrical engineering from Peking University, Beijing, China, in 2014. He is currently pursuing the Ph.D. degree in electrical engineering with the University of California at Davis, Davis.

Mr. Hu was a recipient of the Best Student Paper Award of the 2016 IEEE Radio Frequency Integration Technology Symposium.



Shilei Hao (S'14) received the B.Eng. degree in electronic science and technology from the University of Electronic Science and Technology of China, Chengdu, China, in 2009, and the M.S. degree in microelectronics from Fudan University, Shanghai, China, in 2012. He is currently pursuing the Ph.D. degree in electrical engineering with the University of California at Davis, Davis, CA, USA.



Qun Jane Gu (M'07–SM'15) received the Ph.D. from the University of California at Los Angeles, Los Angeles, in 2007. She is currently an Associate Professor with the University of California at Davis, Davis.